



Introduction & Motivation

Malware continues to pose a significant threat to modern computing systems. With attackers adopting increasingly stealthy techniques, traditional detection methods are becoming less effective. This project aims to study malware from a behavioural perspective — that is, by observing its actions during runtime — to identify patterns and system artefacts that can assist in its detection. The project is motivated by the need to understand how modern malware evades detection and interacts with both Linux and Windows environments. By simulating attacks in a controlled lab setup, we can uncover unique behavioural traits, providing a valuable foundation for improved security monitoring tools.

Methodology

This project was conducted in a controlled, offline malware analysis lab consisting of Kali Linux (attacker machine), Windows 7, and Ubuntu Desktop (victim machines). Virtual machines were created using VMware Workstation Pro, with network isolation enabled to prevent malware from reaching the internet.

Malware samples were executed within these VMs, and system behaviour was monitored using:

- Audit policy configurations and Event Viewer (Windows)
- Script-based logging (Linux)
- Tools like Wireshark and Process Monitor to capture runtime behaviours

Collected data included system artefacts such as process activity, file modifications, registry changes, and network connections. Results were analysed to identify behavioural patterns and evasion techniques used by the malware.

Behavioural Observations

Jigsaw.exe Ransomware Injection Example

- **User Interface Lockdown:** Upon execution, the ransomware launched a full-screen ransom window, threatening file deletion on a countdown timer.
- **File Encryption:** System files, including .jpg, .txt, .mp3, and .pdf, were encrypted and renamed with the .fun extension (e.g., Document.txt.fun).
- **Network Traffic (Wireshark):** The sample generated NetBIOS name queries (NBNS) and local LLMNR traffic but did not attempt outbound connections to external IPs — confirming safe offline behaviour.
- **Registry & Services:** Registry logs captured suspicious persistence indicators, e.g., PowerShell entries and unusual startup paths.
- **Process & Connection Logs:** Runtime logging tools (e.g., custom logging script and netstat) captured:
 - New processes (e.g., PowerShell, procmon.exe)
 - Network services and ports opened by the system during execution
- **Event Viewer:** Key events logged include:
 - 5038 – System integrity changed
 - 7036 – Service state transitions
 - 4616 – Security state changed

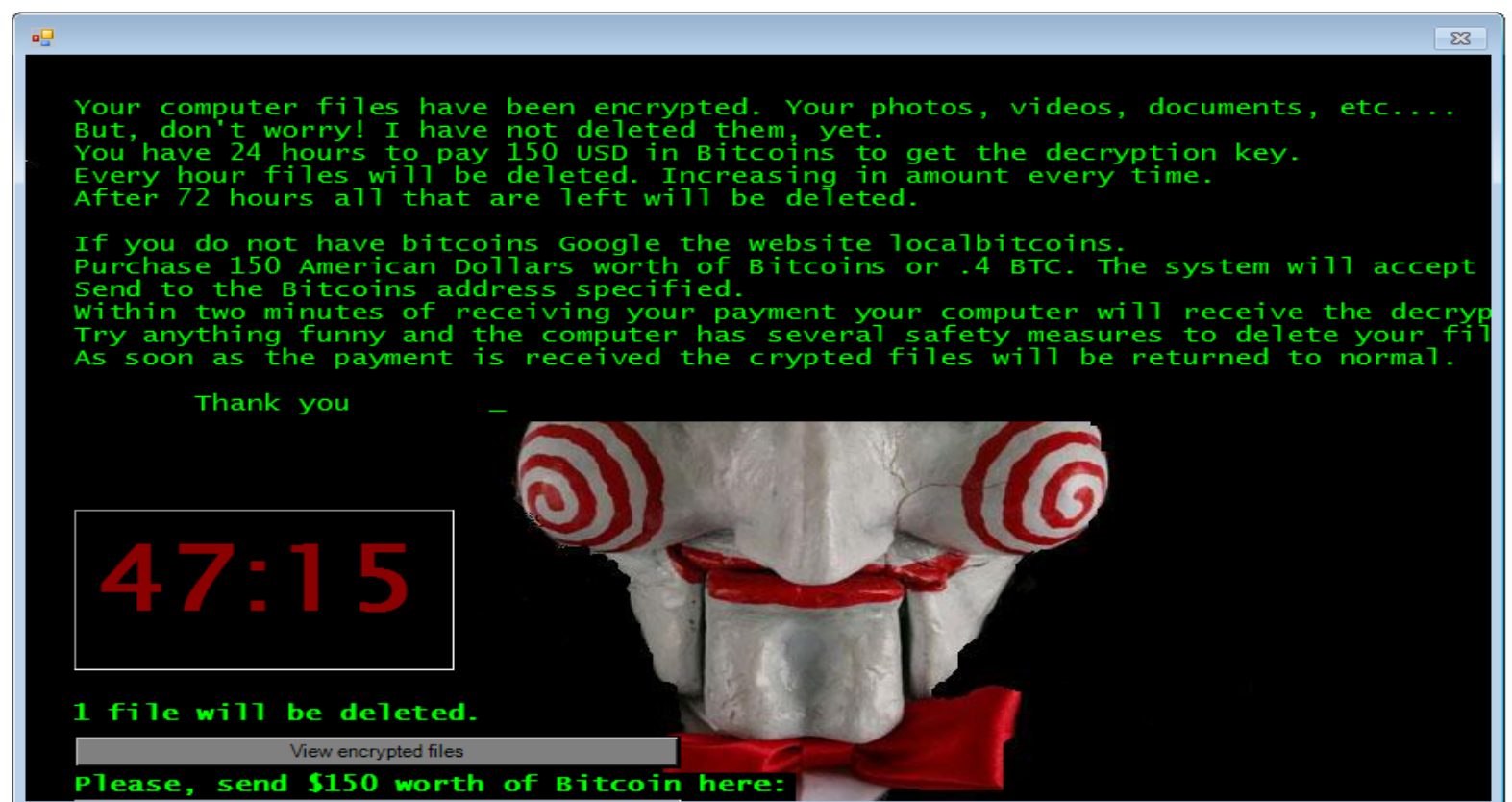


Fig. 1: Jigsaw ransomware window with countdown and Bitcoin demand

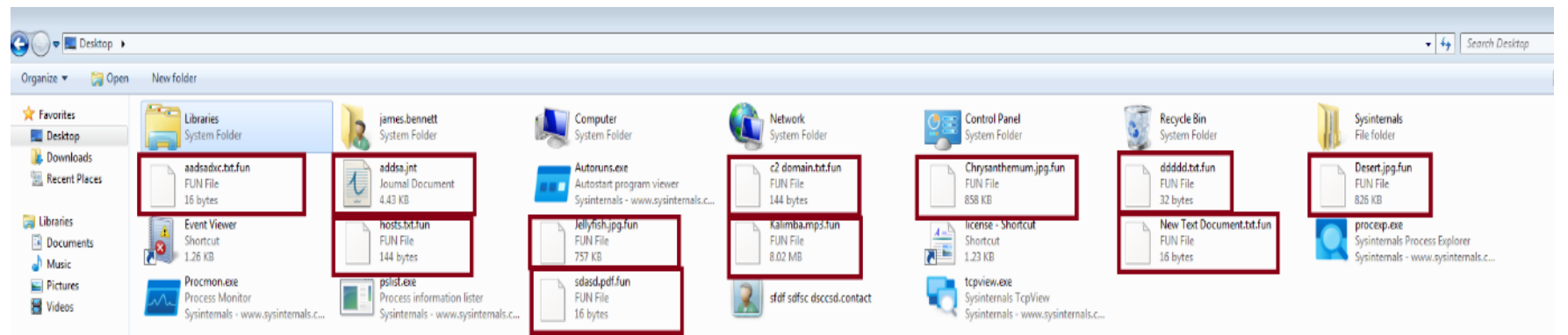


Fig. 2: Encrypted files renamed with .fun extension

Time	Source	Destination	Protocol	Length	Info
132.90.724966	F8B01:1C02:D0F1:ADB...	F8B01:1C02:D0F1:ADB...	LLPMR	84	Standard query Rb482 A upad
132.90.725000	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb482 A upad
134.90.633605	F8B01:1C02:D0F1:ADB...	F8B01:1C02:D0F1:ADB...	LLPMR	84	Standard query Rb482 A upad
136.90.633605	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb482 A upad
136.91.048235	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
137.91.707744	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
138.92.537408	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
138.93.280920	F8B01:1C02:D0F1:ADB...	F8B01:1C02:D0F1:ADB...	LLPMR	84	Standard query Rb482 A upad
140.93.207105	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb482 A upad
141.93.208251	F8B01:1C02:D0F1:ADB...	F8B01:1C02:D0F1:ADB...	LLPMR	84	Standard query Rb482 A upad
142.93.386437	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb482 A upad
143.93.080313	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
144.94.240706	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
145.95.007032	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
146.95.053359	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
147.95.550861	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
148.97.349326	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
149.98.136332	F8B01:1C02:D0F1:ADB...	F8B01:1C02:D0F1:ADB...	LLPMR	84	Standard query Rb482 A upad
150.98.116630	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb482 A upad
151.98.224634	F8B01:1C02:D0F1:ADB...	F8B01:1C02:D0F1:ADB...	LLPMR	84	Standard query Rb482 A upad
152.98.224960	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb482 A upad
153.98.427390	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
154.99.182212	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
155.99.927022	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
156.100.087730	F8B01:1C02:D0F1:ADB...	F8B01:1C02:D0F1:ADB...	LLPMR	84	Standard query Rb482 A upad
157.100.087767	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb482 A upad
158.100.708015	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb482 A upad
160.100.951570	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
161.101.740263	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
162.102.000972	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
163.103.241774	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
164.103.920920	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
165.104.739642	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
166.105.000443	F8B01:1C02:D0F1:ADB...	F8B01:1C02:D0F1:ADB...	LLPMR	84	Standard query Rb482 A upad
167.105.562674	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb482 A upad
168.105.590614	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
169.105.590928	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb482 A upad
170.105.909359	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
171.106.562554	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
172.106.562554	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
173.106.863041	F8B01:1C02:D0F1:ADB...	F8B01:1C02:D0F1:ADB...	LLPMR	84	Standard query Rb47e A upad
174.106.863041	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb47e A upad
175.108.161499	F8B01:1C02:D0F1:ADB...	F8B01:1C02:D0F1:ADB...	LLPMR	84	Standard query Rb47e A upad
176.108.161499	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb47e A upad
177.108.265105	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
178.109.114648	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
179.109.165339	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
180.110.027457	F8B01:1C02:D0F1:ADB...	F8B01:1C02:D0F1:ADB...	LLPMR	84	Standard query Rb47e A upad
181.110.027748	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb47e A upad
182.110.724590	F8B01:1C02:D0F1:ADB...	F8B01:1C02:D0F1:ADB...	LLPMR	84	Standard query Rb47e A upad
183.110.724853	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb47e A upad
184.110.920253	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
185.111.077309	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
186.112.020888	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb47e A upad
187.113.189344	F8B01:1C02:D0F1:ADB...	F8B01:1C02:D0F1:ADB...	LLPMR	84	Standard query Rb47e A upad
188.113.189763	192.168.56.102	224.0.0.252	LLPMR	64	Standard query Rb47e A upad
189.113.287736	F8B01:1C02:D0F1:ADB...	F8B01:1C02:D0F1:ADB...	LLPMR	84	Standard query Rb47e A upad
190.113.288189	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
191.113.481007	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
192.114.048173	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...
193.114.989946	192.168.56.102	192.168.56.255	NBNS	92	Name query NB UPAD-000...

Fig. 3: LLMNR/NetBIOS queries observed after injection

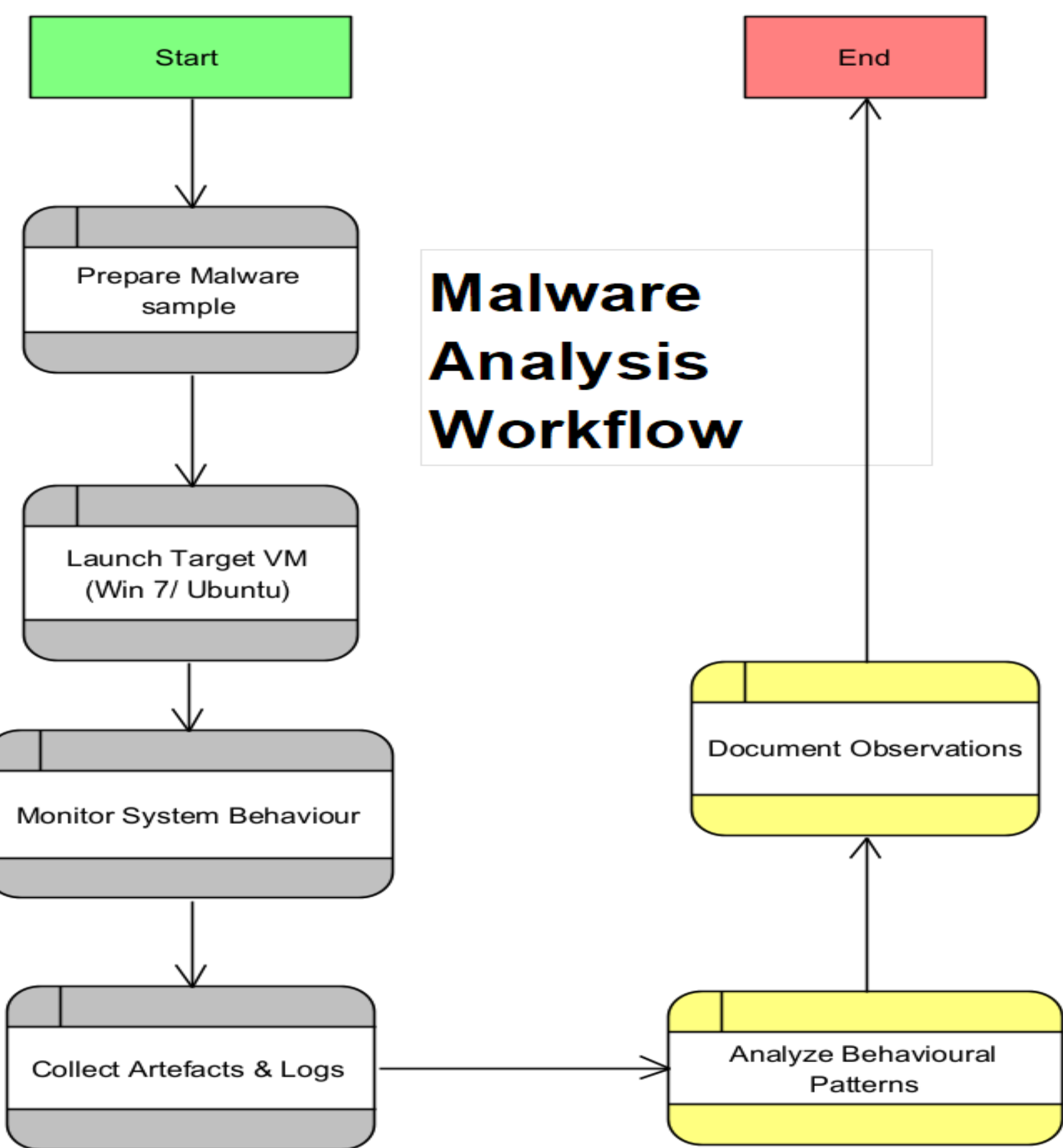
System	Number of events	Source	Date and Time	Event ID	Task Category
System	1	Service Control Manager	4/13/2025 12:03:12 PM	7036	None
System	1	EventLog	4/13/2025 12:03:45 PM	6013	None
System	1	Time Service	4/13/2025 12:05:01 PM	134	None
System	1	Service Control Manager	4/13/2025 12:05:01 PM	7036	None
System	1	Kernel-General	4/13/2025 12:05:01 PM	1	None

Fig. 4: Event Viewer logs showing system service transitions (Event ID 7036) triggered during malware execution.

Level	Date and Time	Source	Event ID	Task Category
Information	4/13/2025 12:02:46 PM	Microsoft Windows security auditing	5038	System Integrity
Information	4/13/2025 12:02:46 PM	Microsoft Windows security auditing	5038	System Integrity
Information	4/13/2025 12:03:01 PM	Microsoft Windows security auditing	4616	Security State Change

Fig. 5: Security audit logs highlighting system integrity changes (Event ID 5038) and a security state change (Event ID 4616).

Malware Analysis Workflow



Conclusions

- Jigsaw ransomware was successfully executed and analyzed in isolated Windows and Linux VMs.
- Key behaviours observed included file encryption, process injection, and service manipulation.
- Artefacts were captured via Event Viewer, Wireshark, and system-level logging scripts.
- Host-only isolation and NAT-based tool access proved effective for secure analysis.
- Behavioural indicators offer a viable foundation for dynamic detection beyond static signatures.

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